

Spyder-notebook

Scope

- Some of you wanted to make notebooks that would have texts, codes, and figures. This document is to help you achieve that goal.
- This document will guide you how to (re-)install Spyder using Miniconda, and how to make a pdf file from Spyder.
- I tried to prepare both Windows and macOS options.

Background

- We installed standalone Spyder, because that's the most simple and straightforward method.
- In order to make a *notebook*, we need to install a plugin named spyder-notebook (<https://docs.spyder-ide.org/current/plugins/notebook.html>).

Installing Spyder again...

- Unfortunately, Spyder installed with standalone installer can't install this plugin. So we need to reinstall Spyder.

Installing the Notebook

If you installed Spyder using conda, the best way to install Spyder-notebook is to run the following command in your terminal or Anaconda prompt on Windows:

```
conda install spyder-notebook -c conda-forge
```

Important

At the moment it is not possible to use this plugin with the Spyder [Standalone installers](#) for Windows and macOS. We're working to make that possible in the future.

Uninstalling Spyder: Windows

Please uninstall your existing Spyder first

- If you're using Windows:

Method 2: Uninstall a manually installed Spyder

1. Find your Spyder installation folder, which is usually in `C:\Program Files` or `C:\Program Files(x86)`.
2. Look for an `uninstall.exe` or similar file and run it to begin the uninstallation. 

Install Miniconda: Windows

Install Miniconda

- I don't like *conda* products, but *pip* option requires more manual labor. So I will compromise and make this demo using Miniconda.
- Go to this [website](#) and check Basic install instructions menu.

Basic install instructions

- Windows installation Click **this** to expand
- macOS/Linux installation
- Verify your install

Windows installation

Installing Miniconda

Copy link

Installing Miniconda (Windows)

Watch on YouTube

1. Download the installer from the Anaconda website or by using your preferred command line interface:

Anaconda website **Command Prompt** PowerShell

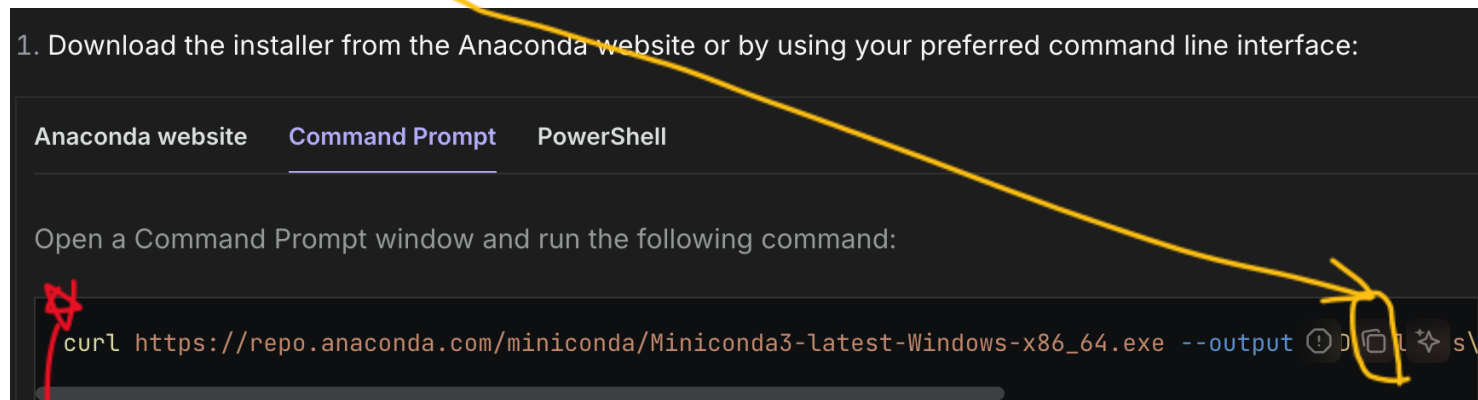
Click **Command Prompt** tab

Open a Command Prompt window and run the following command:

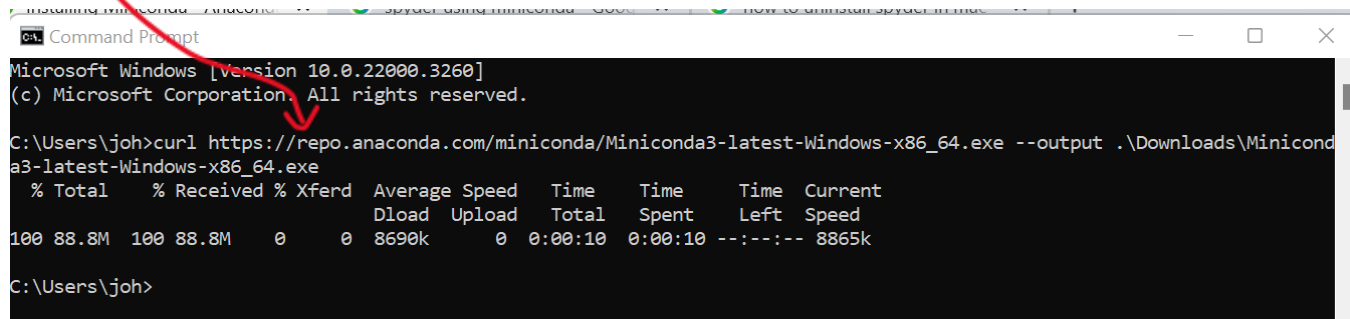
```
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe --output s\
```


Install Miniconda

1. Open Command Prompt
2. Use the button to copy the entire command from the website

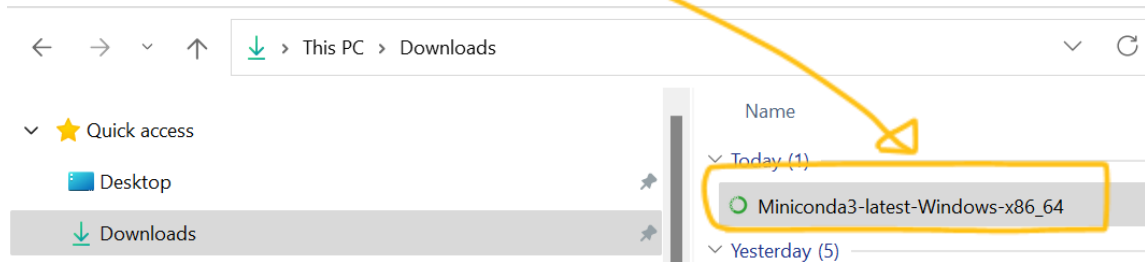


3. Paste the copied command and press Enter in Command Prompt



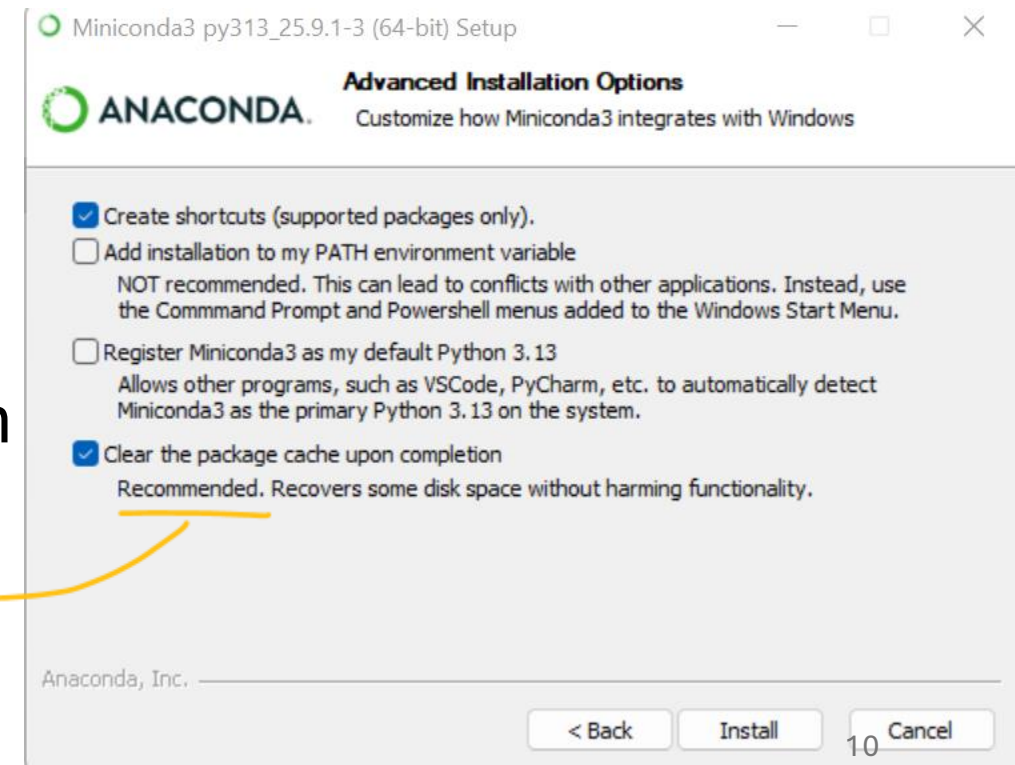
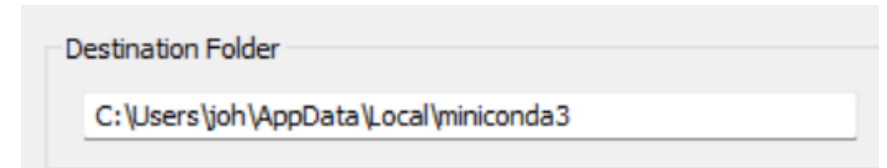
Install Miniconda

- Go to *Downloads* folder and double click the installer.



- As you go through steps, please go with the recommended options:

This is the default destination folder.



Getting started with Conda: Windows

Getting started with Conda

- In the command prompt, type the following:
AppData\Local\miniconda3\Scripts\activate.bat

```
C:\Users\joh>AppData\Local\miniconda3\Scripts\activate.bat  
  
(base) C:\Users\joh>
```

Installing Spyder (with virtual environment)

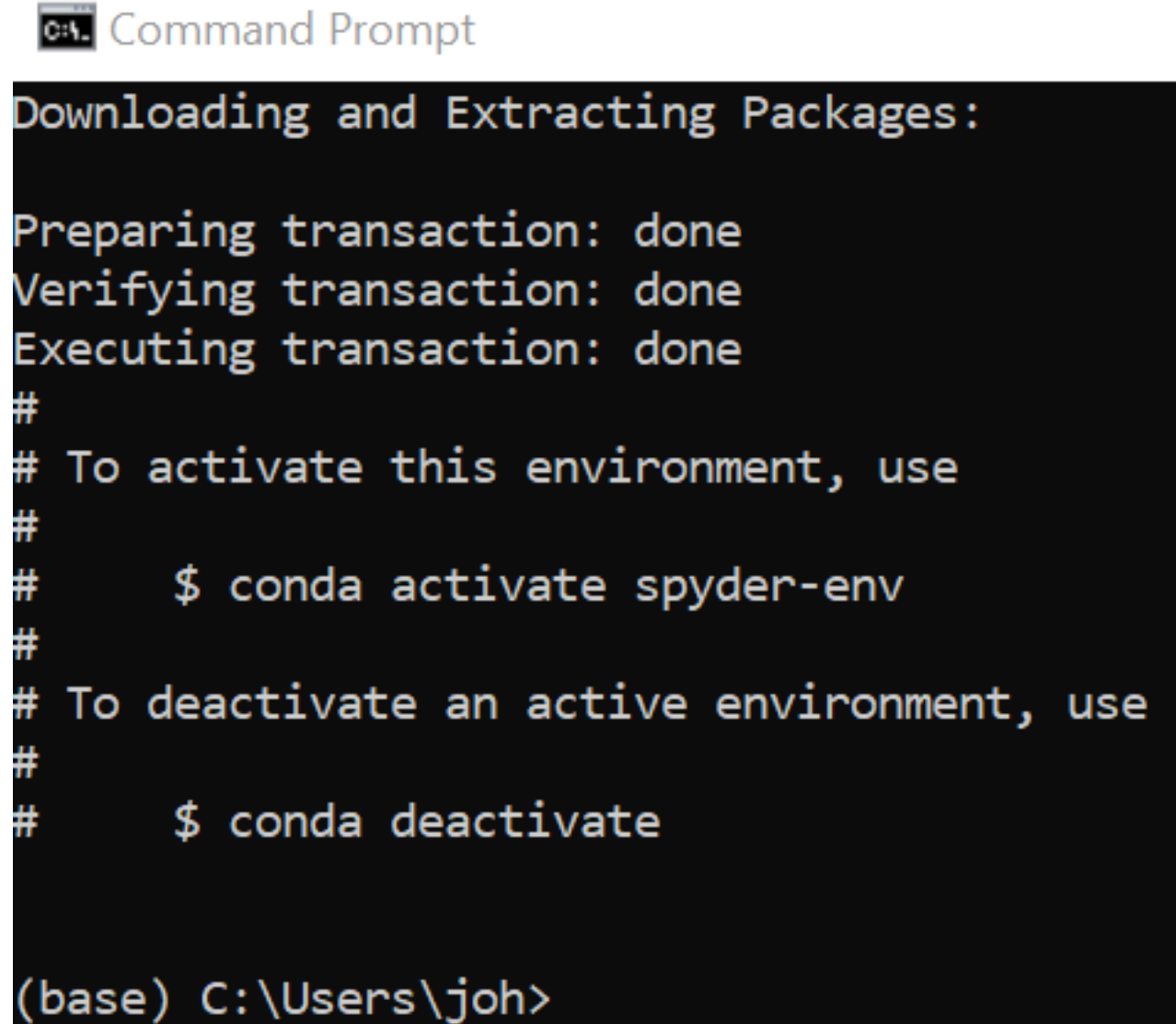
- If you are getting an error message, type the following:
C:\Users\{your_username}\AppData\Local\miniconda3\Scripts\activate.bat
- Then type the following command:
conda create -c conda-forge -n spyder-env spyder

```
(base) C:\Users\joh>conda create -c conda-forge -n spyder-env spyder  
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/main? [(a)ccept/(r)eject/(v)iew]:
```

- Type **a** and press Enter – repeat this 2 more times.

Installing spyder (with virtual environment)

- When asked to proceed, type **y** and press Enter.
- When done, you will see this screen:



```
C:\> Command Prompt

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate spyder-env
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\joh>
```

Launching Spyder: Windows

Move into virtual environment: spyder-env

- As instructed, please type **conda activate spyder-env** and press Enter.

```
(base) C:\Users\joh>conda activate spyder-env  
(spyder-env) C:\Users\joh>
```

- You should see that the *environment* changes from (base) to (spyder-env)

Launch Spyder

- Now type **spyder** and press Enter. This will load Spyder.

```
(spyder-env) C:\Users\joh>spyder
```

- When you start fresh, these are the steps:
 - 1) Open *Command Prompt*.
 - 2) Type **C:\Users\{your_username}\AppData\Local\miniconda3\Scripts\activate.bat** and press Enter.
 - 3) Type **conda activate spyder-env** and press Enter.
 - 4) Type **spyder** and press Enter.

```
Microsoft Windows [Version 10.0.22000.3260]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\joh>AppData\Local\miniconda3\Scripts\activate.bat 1. Activate conda environment
```

```
(base) C:\Users\joh>conda activate spyder-env 2. Activate spyder-env environment
```

```
(spyder-env) C:\Users\joh>spyder 3. Launch Spyder
```

```
(spyder-env) C:\Users\joh>
```



Uninstalling Spyder: macOS

Please uninstall your existing Spyder first

- If you're using macOS:

Manual uninstallation

1. **Open Finder** and navigate to the **Applications** folder.
2. **Find the Spyder application**, then either drag it to the Trash or right-click it and select "Move to Trash".
3. In the Finder menu, click **Go**, then **Go to Folder**.
4. Enter `~/Library` and click **Go**.
5. Go to the `Application Support` folder and delete the `Spyder` folder.
6. Go back to the `~/Library` folder, then open the `caches` folder and delete the `Spyder` folder.
7. Go back to the `~/Library` folder, open the `Preferences` folder, and delete the `Spyder` folder if it exists.
8. Finally, right-click the Trash icon and select **Empty Trash** to permanently delete the files. 

Install Miniconda: macOS

Install Miniconda

- Choose macOS/Linux installation

Basic install instructions

▸ Windows installation

▸ macOS/Linux installation

▸ Verify your install



macOS/Linux installation

ⓘ As of August 15, 2025, Anaconda has stopped building packages for Intel Mac computers (osx-64). Existing Intel (MacOSX-x86_64) installers are still available at <https://repo.anaconda.com/miniconda/> and the last Miniconda installer release for Intel Mac computers will be 25.7.x. For more information, see [our blog on the end of Intel mac support](#).

macOS graphical installer

macOS terminal installer

Linux terminal installer

1. Download the `.sh` installer by opening a terminal and running one of the following commands (depending on your macOS architecture):

Apple Silicon Intel

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
```

Install Miniconda

▼ macOS/Linux installation

❗ As of August 15, 2025, Anaconda has stopped building packages for Intel Mac computers (osx-64). Existing Intel (`MacOSX-x86_64`) installers are still available at <https://repo.anaconda.com/miniconda/> and the last Miniconda installer release for Intel Mac computers will be 25.7.x. For more information, see [our blog on the end of Intel mac support](#).

macOS graphical installer macOS terminal installer Linux terminal installer

1. Download the `.sh` installer by opening a terminal and running one of the following commands (depending on your macOS architecture):

If you're unsure, open terminal and type `uname -a` and check the output. If the last item is `x86_64`, your mac has an Intel chip.

Apple Silicon Intel

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
```

Copy the
command

Install Miniconda

- Open Terminal, paste the command and run:

```
[> curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 112M  100 112M    0     0  7012k      0  0:00:16  0:00:16 --:--:-- 7152k
```

- Run command: **chmod 777 Miniconda3-latest-MacOSX-x86_64.sh**

```
chmod 777 Miniconda3-latest-MacOSX-x86_64.sh
```

- Finally, run **./Miniconda3-latest-MacOSX-x86_64.sh**

```
> ./Miniconda3-latest-MacOSX-x86_64.sh

Welcome to Miniconda3 py313_25.7.0-2

In order to continue the installation process, please review the license
agreement.
Please, press ENTER to continue
>>> |
```


Install Miniconda

- Keep pressing return until you see this screen, where you type **yes** and press return.
- You can install Miniconda3 elsewhere. Here, I will also install it in Downloads (optional).

```
DISCLAIMER: THIS SOFTWARE IS PROVIDED BY ANACONDA  
LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY  
AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL ANACONDA BE LIABLE FOR ANY  
DIRECT, INDIRECT, INCIDENTAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, LOSS OF PROFITS;  
OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN  
CONTRACT, TORT (INCLUDING NEGLIGENCE OR OTHERWISE) OR BY ANY OTHER THEORY OF LIABILITY,  
THE POSSIBILITY OF SUCH DAMAGE.
```

```
[ Do you accept the license terms? [yes|no]  
>>> yes|
```

```
Miniconda3 will now be installed into this location:  
/Users/joh/miniconda3
```

- Press ENTER to confirm the location
- Press CTRL-C to abort the installation
- Or specify a different location below

```
[/Users/joh/miniconda3] >>> /Users/joh/Downloads/miniconda3|
```

Install Miniconda

- You will be asked if you want to update your shell. I would go with no, but you would want to reply yes, as that's recommended.
- Once everything is done, reopen your terminal. Then run **conda list** and see if you see something similar to the right:

```
installation finished.  
Do you wish to update your shell profile to automatically initialize conda?  
This will activate conda on startup and change the command prompt when activated.  
If you'd prefer that conda's base environment not be activated on startup,  
run the following command when conda is activated:  
  
conda config --set auto_activate_base false  
  
You can undo this by running `conda init --reverse $SHELL`? [yes|no]  
[yes] >>> 
```

```
> conda list  
# packages in environment at /Users/joh/Downloads/miniconda3:  
#  
# Name                                Version           Build                Channel  
anaconda-anon-usage                   0.7.2             py313hfb7c958_100  
anaconda-auth                         0.8.6             py313hecd8cb5_0  
anaconda-cli-base                     0.5.2             py313hecd8cb5_0  
annotated-types                       0.6.0             py313hecd8cb5_0  
archspec                             0.2.3             pyhd3eb1b0_0  
boltons                               25.0.0            py313hecd8cb5_0
```

Getting started with Conda: macOS

Getting started with Conda

- From macOS, no need to 'activate'. If you replied 'no' in slide 26, run: **source ~/Downloads/miniconda3/bin/activate**

```
> source ~/Downloads/miniconda3/bin/activate
```

```
~ ▶ |
```

```
◀ ✓ < 4s < base + < 18:58:13
```

If you did not install Miniconda at the default destination (slide 25), this will be:
~/miniconda3/bin/activate

Your terminal may have a different display, but you should see: (base)

- Run the following command: **conda create -c conda-forge -n spyder-env spyder**

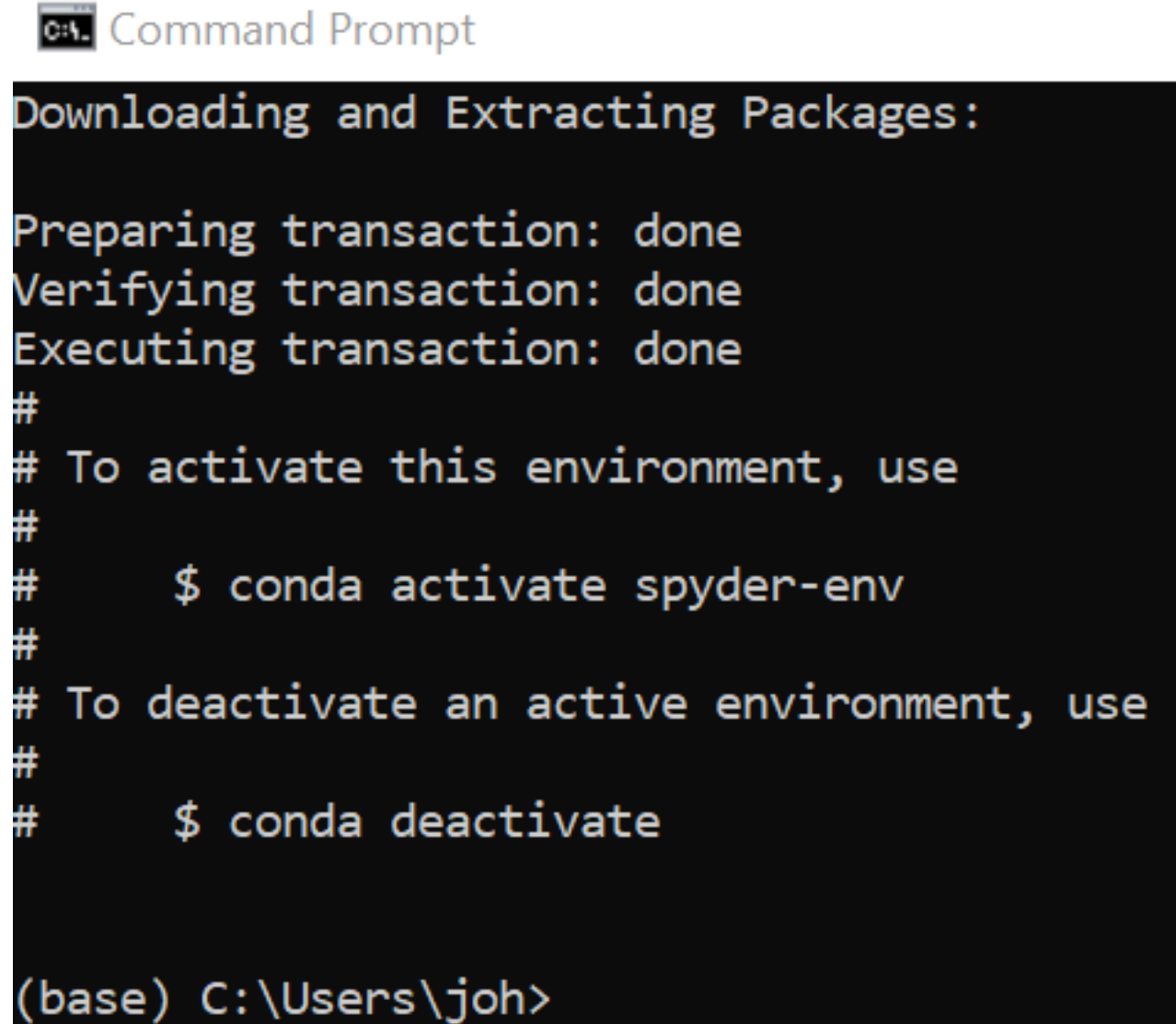
Installing Spyder (with virtual environment)

- In macOS, you may encounter Permission error. If so, run the command using **sudo** (super user do)
- So the command to type is:
sudo conda create -c conda-forge -n spyder-env spyder
Password is equal to the password you use to login.

```
[> sudo conda create -c conda-forge -n spyder-env spyder
[Password:
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/main? [(a)ccept/(r)eject/(v)iew]: a
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/r? [(a)ccept/(r)eject/(v)iew]: a
2 channel Terms of Service accepted
```

Installing spyder (with virtual environment)

- When asked to proceed, type **y** and press Enter.
- When done, you will see this screen:



```
C:\> Command Prompt

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate spyder-env
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\joh>
```

Launching Spyder: macOS

Move into virtual environment: spyder-env

- As instructed, please type **conda activate spyder-env** and press Enter.

A terminal window with a dark background. The first line shows a green prompt character followed by the command 'conda activate spyder-env'. The second line shows a blue prompt character followed by a right arrow and a vertical bar. On the right side of the terminal, there is a status bar with a green checkmark, a left arrow, the text 'spyder-env', a plus sign, a right arrow, and the time '19:05:31'.

```
> conda activate spyder-env
~ ► |
```

- You should see that the *environment* changes from (base) to (spyder-env)

Launch Spyder

- When you start fresh, these are the steps (macOS):
 - 1) Open *Terminal*.
[optional] type command: **source ~/{"folder you installed miniconda"}/miniconda3/bin/activate** and press Return.
 - 2) Type command: **conda activate spyder-env** and press Return.
 - 3) Type command: **spyder** and press Return.

Installing spyder-notebook

Install spyder-notebook

- In the Command Prompt, while you're in (spyder-env), type **conda install spyder-notebook -c conda-forge** and press Enter.

```
(spyder-env) C:\Users\joh>conda install spyder-notebook -c conda-forge
3 channel Terms of Service accepted
Channels:
- conda-forge
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done
```

- When you're on macOS and needs more permission, use **sudo**.
(**sudo conda install spyder-notebook -c conda-forge**)
- When asked to proceed, type **y** and press Enter.

Note: installing other packages

- Run the same command in Command Prompt or Terminal: **conda install {package_name1} {package_name2} ... {package_name N} -c conda-forge**

```
sudo conda install numpy pandas scipy matplotlib seaborn -c conda-forge
```

Run Spyder after installing notebook

- When installation is done, type **spyder** and press Enter.

```
Proceed ([y]/n)? y

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done

(spyder-env) C:\Users\joh>spyder
```

C:\Users\joh\spyder-py3\temp.py

temp.py ×

```
1  # -*- coding: utf-8 -*-
2  """
3  Spyder Editor
4
5  This is a temporary script file.
6  """
7
8
```

Click this button and select "move"

C:\Users\joh



Code not profiled yet

Profile your code to explore which functions and metho...

Help Variable Explorer Debugger Profiler Plots Files

Welcome to
Spyder-
NotebookHere you can open, edit and
create Jupyter notebooks.The items in the *File* menu
act on notebooks if this pane
is active. For example, *File* >
New file... will create a new
notebook.

Console 1/A ×

Python 3.13.9 | packaged by conda-forge | (main, Oct 22 2025,
23:12:41) [MSC v.1944 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.7.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]:

IPython Console History

You can click and drag this pane



Code not profiled yet

Profile your code to explore which functions and metho...

Help Variable Explorer Debugger Profiler Plots Files

Welcome to
Spyder-
Notebook

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The items in the *File* menu
act on notebooks if this pane
is active. For example, *File* >
New file... will create a new

File Edit Search Source Run Debug Consoles Projects Tools Window Help



welcome x untitled0 x

Welcome to Spyder-Notebook

Here you can open, edit and create Jupyter notebooks.

The items in the *File* menu act on notebooks if this pane is active. For example, *File* > *New file...* will create a new notebook.

To resize the text and other elements in a notebook, you can right-click and choose *Zoom in* or *Zoom out*.

2. Once you're locked in,
click this lock icon to fix
the location.

1. Move all the way to the left so that
it will overlap with the editor pane.

Or click this to create a new notebook



Click this tab and you can start editing a Jupyter notebook

Edit View Run Kernel Settings Help Trusted

+ ✂ 📄 📋 ▶ ■ ⚙

Code Python 3 (Spyder) ⌵ ☰

Hey, you made it this far! Congrats!

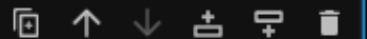
```
[1]: # This is a code cell:  
print(3+4)
```

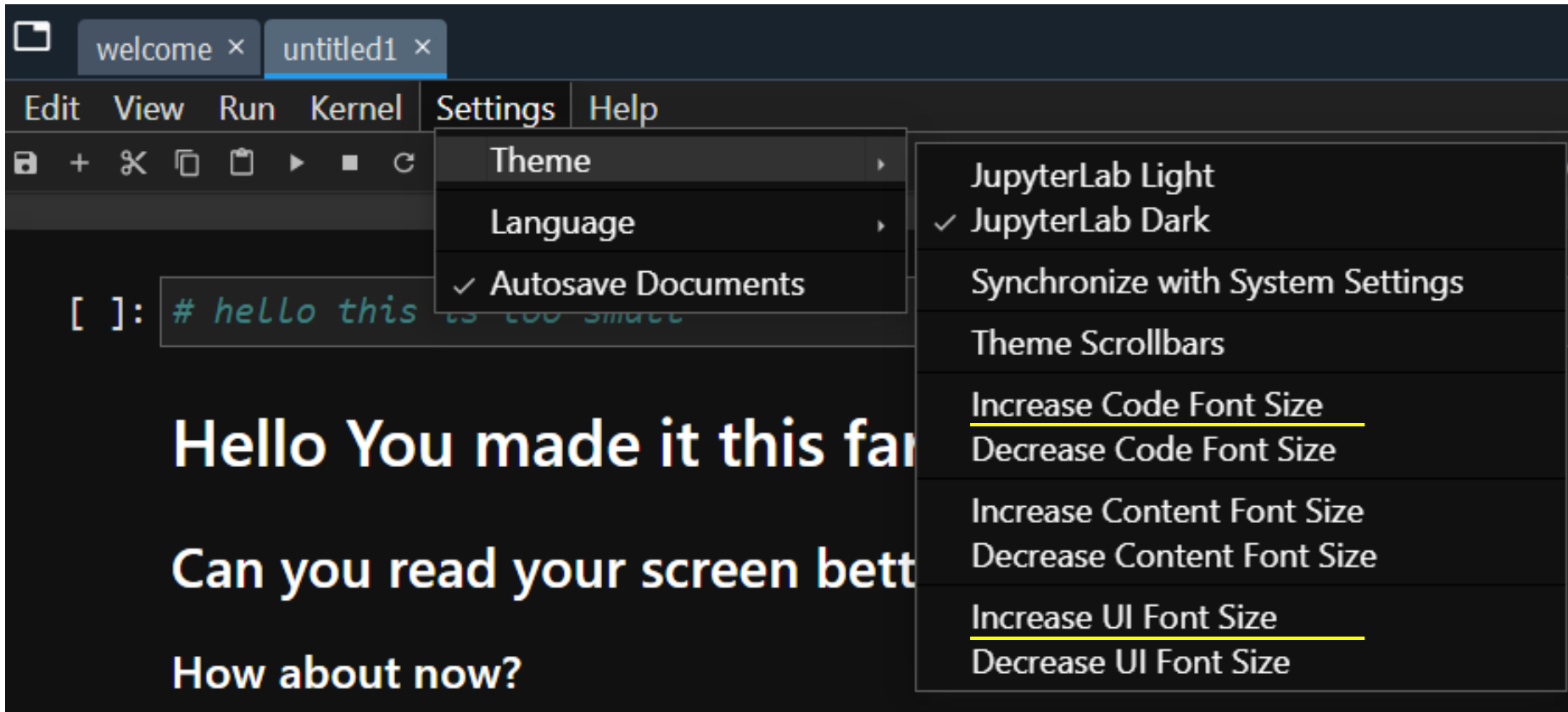
7

```
[2]: # import libraries  
import matplotlib.pyplot as plt  
  
plt.plot([1,2,3,4,5])
```

```
-----  
ModuleNotFoundError                                Traceback (most recent call last)  
Cell In[2], line 2  
      1 # import libraries  
----> 2 import matplotlib.pyplot as plt  
      4 plt.plot([1,2,3,4,5])  
  
ModuleNotFoundError: No module named 'matplotlib'
```

```
[ ]:
```





Use these options to increase font size

This link would be useful to learn more about markdown: [Getting Started | Markdown Guide](#)

Basics

```
# This is the first-level heading
This is a plain text. bold font and italic font are available.

## Second level heading
- bullet point 1
- bullet point 2

### Third level heading
1) numbered item 1
2) numbered item 2

#### Fourth level heading
```
This is how you insert a code snippet

print("Hello World!")
```

Mathematical symbols can be written:  $\alpha = 0.05$ 
```

This is the first-level heading

This is a plain text. **bold** font and *italic* font are available.

Second level heading

- bullet point 1
- bullet point 2

Third level heading

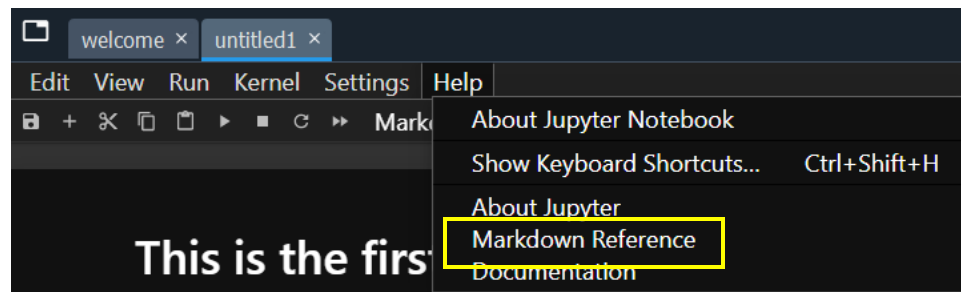
1. numbered item 1
2. numbered item 2

Fourth level heading

```
# This is how you insert a code snippet
```

```
print("Hello World!")
```

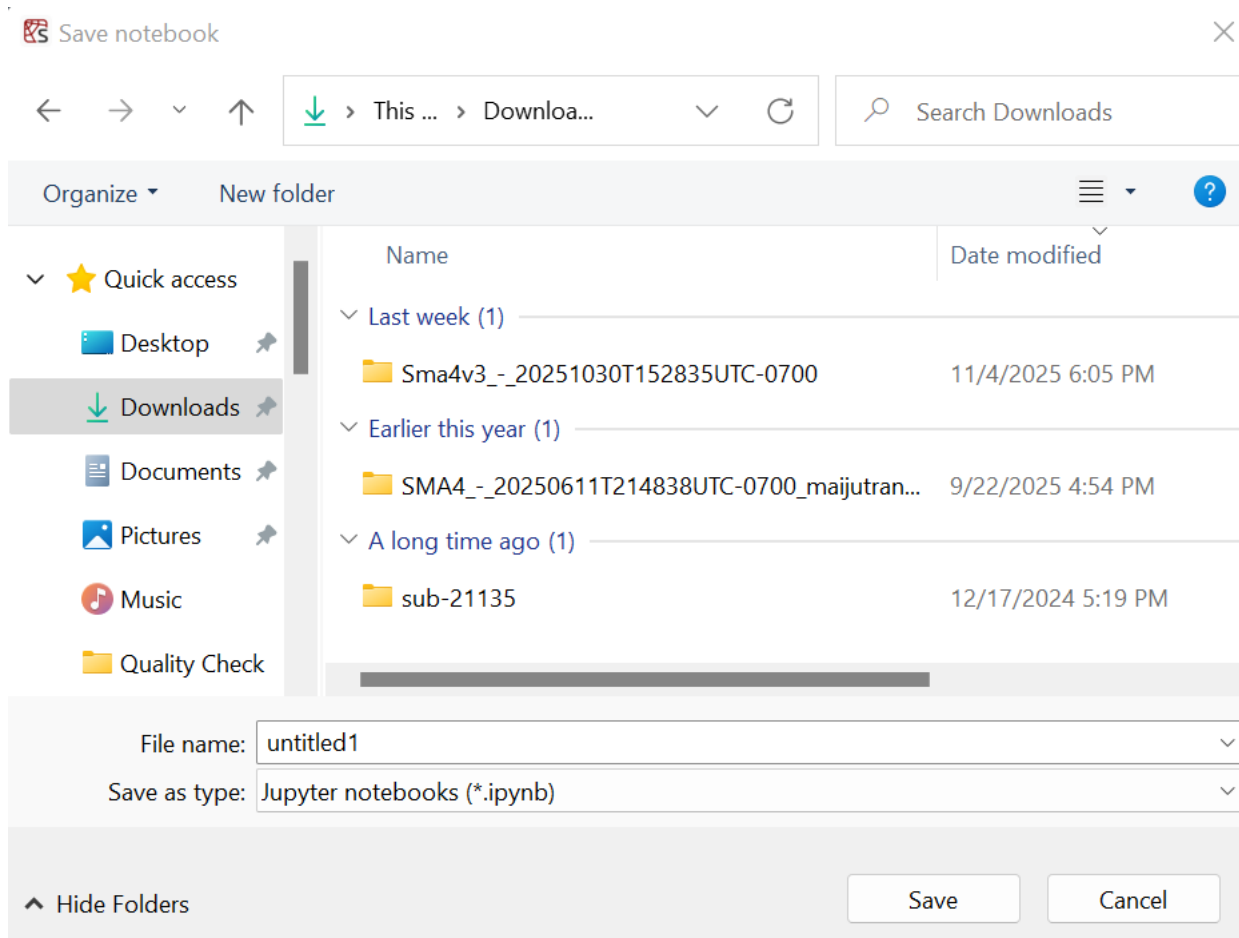
Mathematical symbols can be written: $\alpha = 0.05$



You can read more from here

Export your notebook

- First, save your notebook as a *.ipynb* file.



Export your notebook

- In Command Prompt, move to the folder where you saved your .ipynb. Use **cd {destination directory}** command.

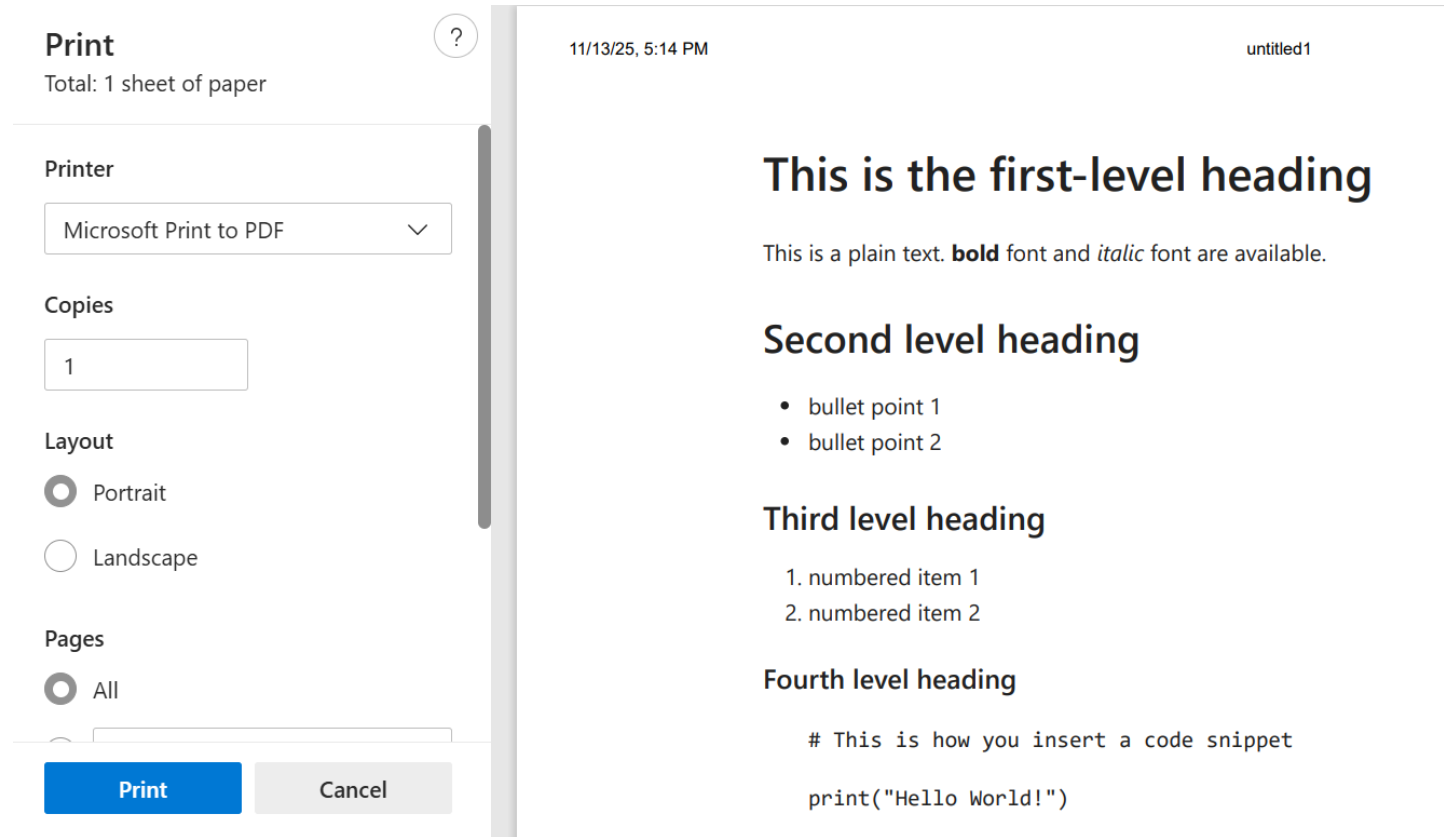
```
(spyder-env) C:\Users\joh>cd Downloads  
(spyder-env) C:\Users\joh\Downloads>
```

- Then type **jupyter nbconvert --to html {name of your ipynb file}** and press Enter

```
(spyder-env) C:\Users\joh\Downloads>jupyter nbconvert --to html untitled1.ipynb  
[NbConvertApp] Converting notebook untitled1.ipynb to html  
[NbConvertApp] Writing 271021 bytes to untitled1.html
```

Export your notebook

- Open your html file using any browser (Chrome, Firefox, Edge...).
- Then print to a pdf file.



Export your notebook

- Open your html file using any browser (Chrome, Firefox, Edge...).
- Then print to a pdf file.

